# Type MS · Describe how an impulse crosses a synapse

## Answer 1 of 10 · Question 3 part a.iii · 6 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (a) | (i) | Synapse/synaptic cleft; | | 1 |
| | (ii) | arrow pointing from top to bottom; | | 1 |
| | (iii) | any six from<br>• vesicles;<br>• contain neurotransmitter/ named neurotransmitter;<br>• (vesicles)fuse/binds/attaches with pre-synaptic membrane/( vesicles) moves to pre-synaptic membrane;<br>• exocytosis/release of neurotransmitter;<br>• which diffuses (across the gap/synapse)/description of diffusion;<br>• binds/attaches with receptors on post-synaptic membrane;<br>• (initiates)new impulse; | | 6 |
| | (iv) | • neurotransmitter only released from pre-synaptic membrane/ neurone A/first neurone/(vesicles) with neurotransmitter in pre-synaptic knob/neurone A/first neurone;<br>• diffuses down a concentration gradient/description of concentration gradient;<br>• receptors only on post-synaptic membrane/neurone B/ second neurone; | | 3 |
| (b) | | any two from<br>• addiction;<br>• stroke/heart problems/high blood pressure;<br>• steal to fund habit;<br>• anxiety;<br>• depression;<br>• hallucinations;<br>• memory loss;<br>• brain damage/ damage nerve cells;<br>• paranoia;<br>• overdose/death; | | 2 |

**Total 13 marks**

## Type MS · Describe how an impulse crosses a synapse

### Answer 1 of 10 · Question 3 part a.iii · 6 marks

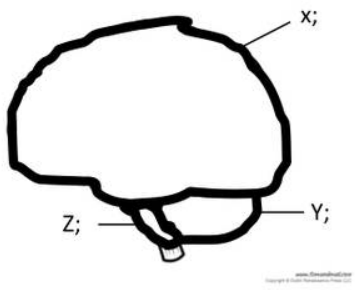
2024 NOV · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Describe how an impulse crosses a synapse

### Answer 2 of 10 · Question 7 part b.iii · 3 marks

2019 JUN · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 7 (a) |  | Letters can be anywhere within outline shown.<br>Allow labels in place of than letters. | 3 |
| (b) (i) | Motor; | | 1 |
| (ii) | Any one from the following:<br><br>(sensory neurone):<br>longer dendrites/shorter axon/cell body outside of spinal cord/CNS/in middle of neurone/transmit impulse to the CNS/brain/spinal cord<br><br>(motor neurone)<br>shorter dendrites/longer axon/cell body at one end of neurone/inside spinal cord/transmit impulses from CNS/brain/spinal cord | Answer must be comparative and state which type of neurone is being referred to | 1 |
| (iii) | • chemicals/neurotransmitters;<br>• diffuse;<br>• across a synapse;<br>• bind to receptors (on post-synaptic membrane); | | 3 |
| (c) | • stem cells are undifferentiated/unspecialised/can differentiate;<br>• could be made to form nerve cells;<br>• to replace damaged cells in brain; | | 1<br>1<br>1 |

Total question 7 = 11 marks

## Type MS · Describe how an impulse crosses a synapse

### Answer 2 of 10 · Question 7 part b.iii · 3 marks

2019 JUN · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Describe how an impulse crosses a synapse

### Answer 3 of 10 · Question 5 part c.ii · 3 marks

2020 NOV · Paper 01 · extract 1 of 2 · find part c.ii in this mark scheme extract

| Number | | | |
|---|---|---|---|
| 5 (a) | any three from<br>• rapid response (to a stimulus) (1)<br>• involuntary/automatic (1)<br>• no brain involvement (1)<br>• protective (1) | | 3 max |
| (b) | • A = motor (1)<br>• B = relay/association/connector (1)<br>• C = sensory (1) | | 3 |
| (c) (i) | • fewer branches/dendrons/dendrites (1)<br>• shorter branches/ dendrons/dendrites (1)<br>• fewer knobs/synapses (1) | R axons | 3 |
| (ii) | • transmits impulse (1)<br>• across synapse/(synaptic) cleft (1)<br>• stimulates impulse in next neurone (1) | | 3 |
| (iii) | • secreted by synapses/synaptic knob (1)<br>• fewer synapses/knobs (1)<br>• less secreted by remaining synapses/knobs (1) | | 3 |

Total 15 marks

## Type MS · Describe how an impulse crosses a synapse

### Answer 3 of 10 · Question 5 part c.ii · 3 marks

2020 NOV · Paper 01 · extract 2 of 2 · find part c.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

## Type MS · Describe how an impulse crosses a synapse

### Answer 4 of 10 · Question 8 part c · 6 marks

2021 JUN · Paper 01 · find part c in this mark scheme extract

| number | | | | |
|---|---|---|---|---|
| 8 | (a) | any three from<br>• rapid response;<br>• automatic;<br>• to protect body;<br>• from harmful effects/named example; | | 3 |
| | (b) | correct position; | | 1 |
| | (c) | any six from<br>• impulse arrives at end of axon;<br>• synapse;<br>• causes release of neurotransmitter;<br>• e.g. acetylcholine;<br>• from vesicles;<br>• diffuses across gap/cleft;<br>• combines with/stimulates (post synaptic) receptors;<br>• triggers impulse; | | 6 |

Total 10 marks

## Type MS · Describe how an impulse crosses a synapse

### Answer 5 of 10 · Question 6 part b.ii · 4 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 6 | (a) | <table><tr><th>Nervous system</th><th>Hormonal system</th></tr><tr><td>nerve fibres/cells;</td><td>blood;</td></tr><tr><td>electrical/electrochemical;</td><td>chemical;</td></tr><tr><td>fast;</td><td>slow;</td></tr></table> | Allow nerves / neurones<br>Reject hormone | 3 |
| | (b) (i) | arrow drawn left to right; | | 1 |
| | (ii) | • neurotransmitter;<br>• released from synapse/synaptic knobs;<br>• diffuses across gap;<br>• bind to receptors on next cell; | | 4 |
| | (iii) | • neurotransmitter only released at one side of synapse;<br>• can only diffuse in one direction;<br>• from high to low concentration;<br>• receptors only present on post synaptic membrane; | | 4 |

Total 12 marks

## Type MS · Describe how an impulse crosses a synapse

### Answer 5 of 10 · Question 6 part b.ii · 4 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Describe how an impulse crosses a synapse

Answer 6 of 10 · Question 3 part b.i · 1 mark

2022 JUN · Paper 02 · find part b.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) | In the following order:<br>• receptors;<br>• sensory;<br>• relay;<br>• motor; | | 4 |
| (b) (i) | synapse; | | 1 |
| (ii) | • chemicals/neurotransmitters;<br>• diffuse across synaptic clef/gap; | | 2 |
| (c) | • 60 × 5;<br>• 300 x 120 = 36 000; | | 2 |
| | | | Total 9 |

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Describe how an impulse crosses a synapse

Answer 7 of 10 · Question 3 part b.ii · 2 marks

2022 JUN · Paper 02 · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) | In the following order:<br>• receptors;<br>• sensory;<br>• relay;<br>• motor; | | 4 |
| (b) (i) | synapse; | | 1 |
| (ii) | • chemicals/neurotransmitters;<br>• diffuse across synaptic clef/gap; | | 2 |
| (c) | • 60 × 5;<br>• 300 x 120 = 36 000; | | 2 |
| | | | Total 9 |

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Describe how an impulse crosses a synapse

### Answer 8 of 10 · Question 5 part a.i · 2 marks

2023 JUN · Paper 02 (regional R) · find part a.i in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 5 | (a) (i) | Any two from:<br>• chemical;<br>• diffused from one neurone to another;<br>• allows neurones to communicate / reference to communication between presynaptic and post synaptic neurones; | | 2 |
| | (ii) | • neurotransmitter/dopamine/ diffuses/travels across synapse;<br>• binds with receptor (on postsynaptic membrane) | | 2 |
| | (iii) | Any two from:<br>• increased movement;<br>• increased heart rate/blood pressure;<br>• more energy;<br>• sleeping difficulties;<br>• anxiety/stress;<br>• hallucinations; | | 2 |
| | (b) (i) | Any two from<br>• tremors/random muscular movement / involuntary movement;<br>• slower movement;<br>• speech/writing changes;<br>• impaired posture/balance;<br>• sleeping problems;<br>• rigidity/stiffness;<br>• anxiety/depression/mental health/memory problems; | | 2 |
| | (ii) | • (145 000 ÷ 100) (x 18) = 26 100;<br>• (26 100 + 145 000) = 171 100; | Allow ecf for mp2 - if mp1 has an incorrect value but this value is added to 145 000 to arrive at the correct answer for the calculation award 1 mark<br><br>Full marks for correct final answer | 2 |

**Total for question 5 = 10 marks**

## Type MS · Describe how an impulse crosses a synapse

### Answer 9 of 10 · Question 5 part a.ii · 2 marks

2023 JUN · Paper 02 (regional R) · find part a.ii in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 5 | (a) (i) | Any two from:<br>• chemical;<br>• diffused from one neurone to another;<br>• allows neurones to communicate / reference to communication between presynaptic and post synaptic neurones; | | 2 |
| | (ii) | • neurotransmitter/dopamine/ diffuses/travels across synapse;<br>• binds with receptor (on postsynaptic membrane) | | 2 |
| | (iii) | Any two from:<br>• increased movement;<br>• increased heart rate/blood pressure;<br>• more energy;<br>• sleeping difficulties;<br>• anxiety/stress;<br>• hallucinations; | | 2 |
| | (b) (i) | Any two from<br>• tremors/random muscular movement / involuntary movement;<br>• slower movement;<br>• speech/writing changes;<br>• impaired posture/balance;<br>• sleeping problems;<br>• rigidity/stiffness;<br>• anxiety/depression/mental health/memory problems; | | 2 |
| | (ii) | • (145 000 ÷ 100) (x 18) = 26 100;<br>• (26 100 + 145 000) = 171 100; | Allow ecf for mp2 - if mp1 has an incorrect value but this value is added to 145 000 to arrive at the correct answer for the calculation award 1 mark<br><br>Full marks for correct final answer | 2 |

**Total for question 5 = 10 marks**

## Type MS · Describe how an impulse crosses a synapse

Answer 10 of 10 · Question 3 part a.i · 1 mark

2024 NOV · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

| | | | | | |
|---|---|---|---|---|---|
| 3 | (a) | (i) | Synapse/synaptic cleft; | | 1 |
| | | (ii) | arrow pointing from top to bottom; | | 1 |
| | | (iii) | any six from<br>• vesicles;<br>• contain neurotransmitter/ named neurotransmitter;<br>• (vesicles)fuse/binds/attaches with pre-synaptic membrane/( vesicles) moves to pre-synaptic membrane;<br>• exocytosis/release of neurotransmitter;<br>• which diffuses (across the gap/synapse)/description of diffusion;<br>• binds/attaches with receptors on post-synaptic membrane;<br>• (initiates)new impulse; | | 6 |
| | | (iv) | • neurotransmitter only released from pre-synaptic membrane/ neurone A/first neurone/(vesicles) with neurotransmitter in pre-synaptic knob/neurone A/first neurone;<br>• diffuses down a concentration gradient/description of concentration gradient;<br>• receptors only on post-synaptic membrane/neurone B/ second neurone; | | 3 |
| | (b) | | any two from<br>• addiction;<br>• stroke/heart problems/high blood pressure;<br>• steal to fund habit;<br>• anxiety;<br>• depression;<br>• hallucinations;<br>• memory loss;<br>• brain damage/ damage nerve cells;<br>• paranoia;<br>• overdose/death; | | 2 |

**Total 13 marks**

## Type MS · Describe how an impulse crosses a synapse

### Answer 10 of 10 · Question 3 part a.i · 1 mark

2024 NOV · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|